Lab Exercise 18- Volume in Kubernetes Cluster

Volumes in Kubernetes provide a way to store and share data among containers in a Pod. This lab exercise will guide you through the creation and use of different types of volumes in Kubernetes.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

```
C:\Program Files\WindowsAp
PowerShell 7.5.4
PS C:\Windows\System32> cd C:\Users\HP\Desktop\COAS_Lab\Lab_8
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 2: Create a Pod with an emptyDir Volume

Create a file named **emptydir-pod.yaml** with the following content:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-vol1
spec:
  containers:
    - image: nginx
      name: test-container
      volumeMounts:
        - mountPath: /data
          name: first-volume
  volumes:
    - name: first-volume
      emptyDir: {}
```

In this manifest:

The emptyDir volume is a temporary storage that is created when the Pod is assigned to a node and exists as long as the Pod is running.

Apply the manifest to create the Pod:

```
kubectl apply -f emptydir-pod.yaml
```

emptydir-pod.yaml

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-vol1
spec:
  containers:
    - image: nginx
      name: test-container
      volumeMounts:
        - mountPath: /data
          name: first-volume
  volumes:
    - name: first-volume
      emptyDir: {}
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f emptydir-pod.yaml
pod/test-vol1 created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 3: Verify the Pod and Volume

Check the status of the Pod:

```
kubectl get pods
```

Access the Pod's shell:

```
kubectl exec -it test-vol1 -- bash
```

Inside the Pod, create a file in the /data directory:

```
touch f1 f2 f3
```

You should see the text "Hello, Kubernetes!".

Exit the Pod shell:

```
exit
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
test-vol1   1/1     Running   0          78s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
test-vol1   1/1     Running   0          80s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl exec -it test-vol1 -- bash
root@test-vol1:/# touch f1 f2 f3
root@test-vol1:/# exit
exit
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 4: Create a Pod with a hostPath Volume

Create a file named **hostpath-pod.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: test-vol2
spec:
  containers:
    - image: coolgourav147/nginx-custom
      name: test-container
      volumeMounts:
        - mountPath: /data
          name: first-volume
  volumes:
    - name: first-volume
      hostPath:
        path: /test
```

In this manifest:

The hostPath volume mounts a directory from the host into the Pod at /mnt/data.

The busybox container writes the current date to a log file every 5 seconds.

Apply the manifest to create the Pod:

```
kubectl apply -f hostpath-pod.yaml
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-vol2
spec:
  containers:
    - image: coolgourav147/nginx-custom
      name: test-container
      volumeMounts:
        - mountPath: /data
          name: first-volume
  volumes:
    - name: first-volume
      hostPath:
        path: /test
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f hostpath-pod.yaml
pod/test-vol2 created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 5: Verify the Pod and Volume

Check the status of the Pod:

```
kubectl get pods
```

Access the Pod's logs to see the data being written:

```
kubectl exec -it test-vol2 -- bash
```

You should see the dates being appended to the log file.

Check the host directory to verify the volume:

If you have access to the node running the Pod, you can verify the data in /tmp/data/log.txt.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME        READY   STATUS              RESTARTS   AGE
test-vol1   1/1     Running             0          6m3s
test-vol2   0/1     ContainerCreating   0          2m47s
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
test-vol1   1/1     Running   0          17m
test-vol2   1/1     Running   0          14m
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl exec -it test-vol2 -- bash
root@test-vol2:/# cd /data
root@test-vol2:/data# ls
log.txt
root@test-vol2:/data# cat log.txt
Tue Mar  3 09:51:33 UTC 2026
Tue Mar  3 09:51:38 UTC 2026
Tue Mar  3 09:51:43 UTC 2026
Tue Mar  3 09:51:48 UTC 2026
Tue Mar  3 09:51:53 UTC 2026
Tue Mar  3 09:51:58 UTC 2026
Tue Mar  3 09:52:03 UTC 2026
Tue Mar  3 09:52:08 UTC 2026
Tue Mar  3 09:52:13 UTC 2026
Tue Mar  3 09:52:18 UTC 2026
Tue Mar  3 09:52:23 UTC 2026
Tue Mar  3 09:52:28 UTC 2026
Tue Mar  3 09:52:33 UTC 2026
Tue Mar  3 09:52:38 UTC 2026
Tue Mar  3 09:52:43 UTC 2026
Tue Mar  3 09:52:48 UTC 2026
Tue Mar  3 09:52:53 UTC 2026
Tue Mar  3 09:52:58 UTC 2026
Tue Mar  3 09:53:03 UTC 2026
Tue Mar  3 09:53:45 UTC 2026
Tue Mar  3 09:53:50 UTC 2026
Tue Mar  3 09:53:55 UTC 2026
Tue Mar  3 09:54:00 UTC 2026
Tue Mar  3 09:54:05 UTC 2026
root@test-vol2:/data#
```

```
root@test-vol2:/data# exit
exit
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl delete pod test-vol2
pod "test-vol2" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f hostpath-pod.yaml
pod/test-vol2 created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
test-vol1   1/1     Running   0          24m
test-vol2   1/1     Running   0          11s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl exec -it test-vol2 -- bash
root@test-vol2:/# cd /data
root@test-vol2:/data# cat log.txt
Tue Mar  3 09:51:33 UTC 2026
Tue Mar  3 09:51:38 UTC 2026
Tue Mar  3 09:51:43 UTC 2026
Tue Mar  3 09:51:48 UTC 2026
Tue Mar  3 09:51:53 UTC 2026
Tue Mar  3 09:51:58 UTC 2026
Tue Mar  3 09:52:03 UTC 2026
Tue Mar  3 09:52:08 UTC 2026
Tue Mar  3 09:52:13 UTC 2026
Tue Mar  3 09:52:18 UTC 2026
Tue Mar  3 09:52:23 UTC 2026
Tue Mar  3 09:52:28 UTC 2026
Tue Mar  3 09:52:33 UTC 2026
Tue Mar  3 09:52:38 UTC 2026
Tue Mar  3 09:52:43 UTC 2026
Tue Mar  3 09:52:48 UTC 2026
Tue Mar  3 09:52:53 UTC 2026
Tue Mar  3 09:52:58 UTC 2026
Tue Mar  3 09:53:03 UTC 2026
Tue Mar  3 09:53:45 UTC 2026
Tue Mar  3 09:53:50 UTC 2026
Tue Mar  3 09:53:55 UTC 2026
Tue Mar  3 09:54:00 UTC 2026
Tue Mar  3 09:54:05 UTC 2026
root@test-vol2:/data#
```